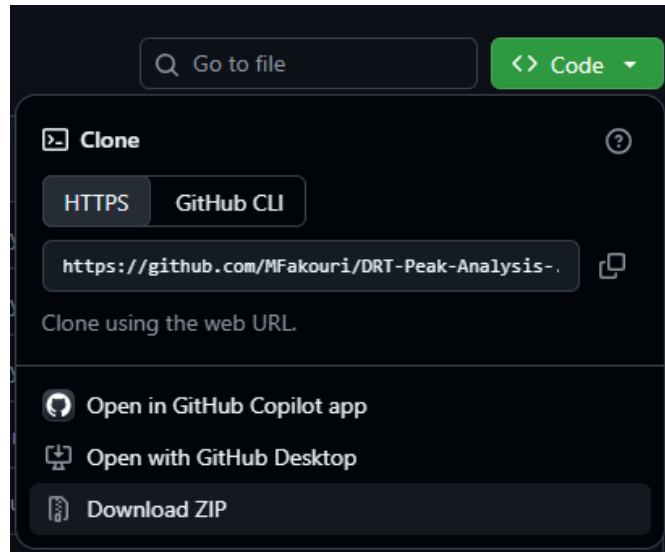


DRT Peak Analysis User Guide

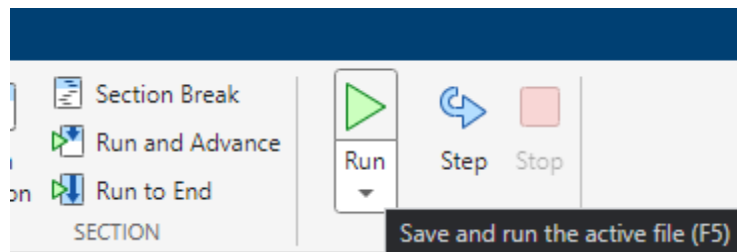
1. Download the last version from the link below and extract the compressed file.

<https://github.com/MFakouri/DRT-Peak-Analysis->



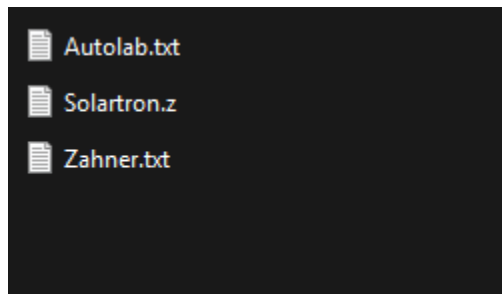
2. Run the MATLAB file DRTtools.m.

docs	2026-08-15 8:56 PM	File folder	
import file samples	2026-08-15 8:44 PM	File folder	
src	2026-08-15 8:14 PM	File folder	
DRTtools.fig	2022-05-23 1:46 AM	MATLAB Figure	70 KB
DRTtools.m	2026-08-15 9:19 PM	MATLAB Code	80 KB

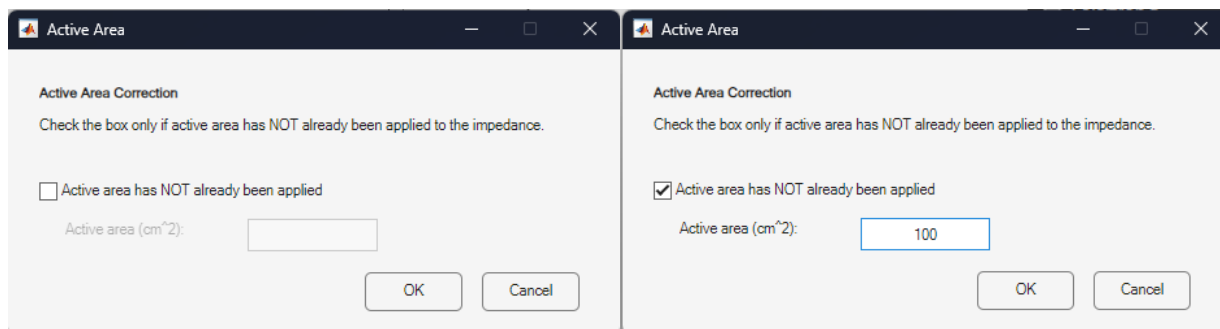


3. Import EIS file. TXT and CSV files exported from different instruments, including Solartron, Autolab, Gamry, Admiral, and Zahner can be used without any modification.

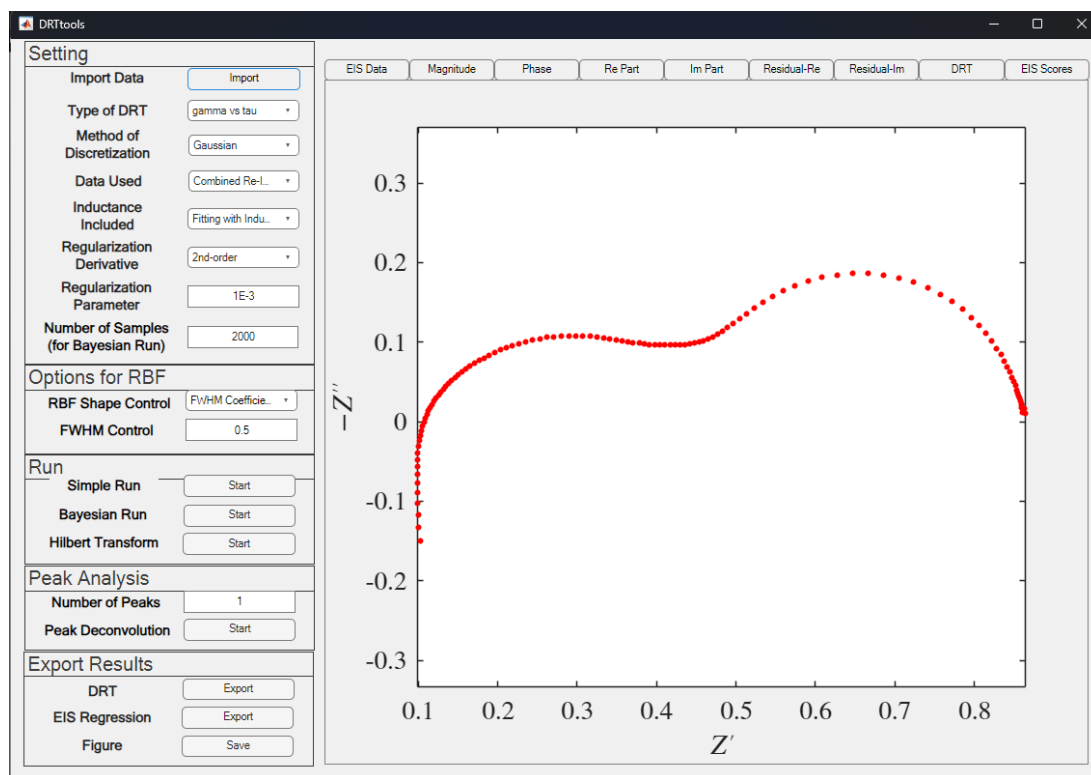
Setting	
Import Data	Import
Type of DRT	gamma vs tau
Method of Discretization	Gaussian
Data Used	Combined Re-I...
Inductance Included	Fitting with Indu...
Regularization Derivative	2nd-order
Regularization Parameter	1E-3
Number of Samples (for Bayesian Run)	2000
Options for RBF	
RBF Shape Control	FWHM Coefficie...
FWHM Control	0.5



4. If the polarization data are already in $\Omega \cdot \text{cm}^2$, leave the checkbox unchecked and click **OK**. If the data are in Ω , check the box and enter the active area.

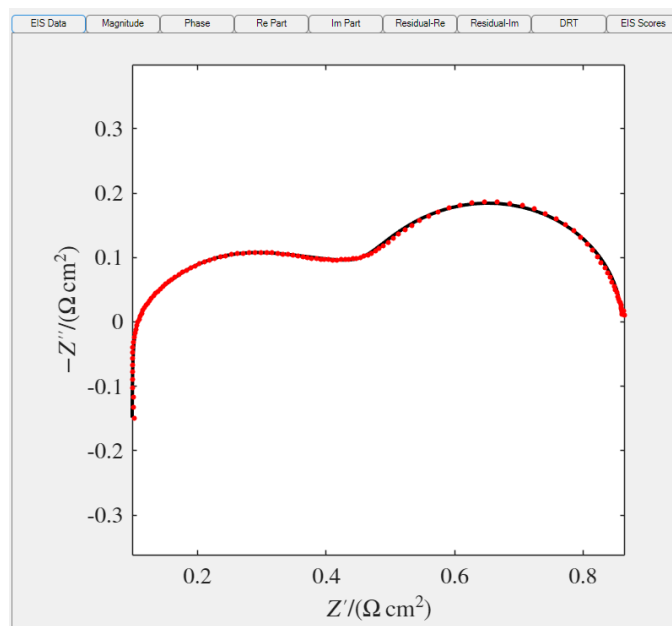


5. If the Nyquist plot is displayed correctly, the data have been imported successfully.

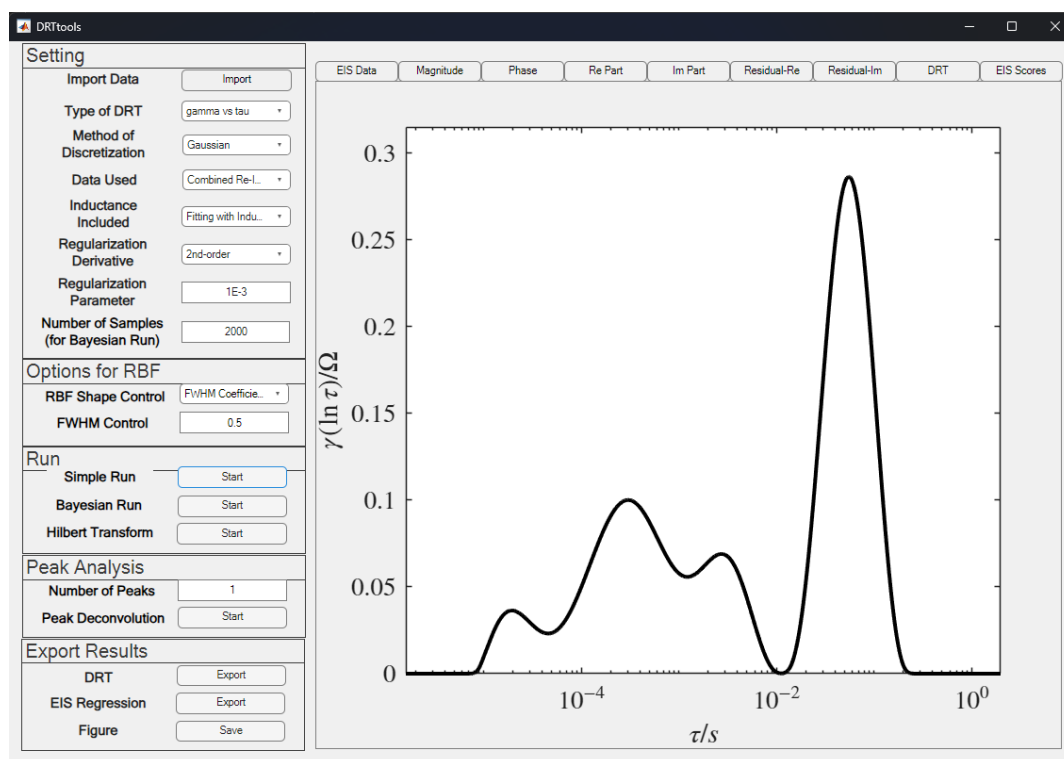


6. Click “Simple Run” to start the DRT analysis. After the Simple Run is completed, check the “EIS Data” window to make sure the fitted curve agrees well with the experimental data.

Adjust the parameters, particularly the Regularization Parameter and FWHM Control, and Click “Simple Run” again, to obtain the best fit in the “EIS Data” window. If a good fit cannot be achieved, especially at very high or very low frequencies, the EIS data may not be sufficiently reliable, and the experiment should be repeated after appropriate modifications to the test conditions or setup.



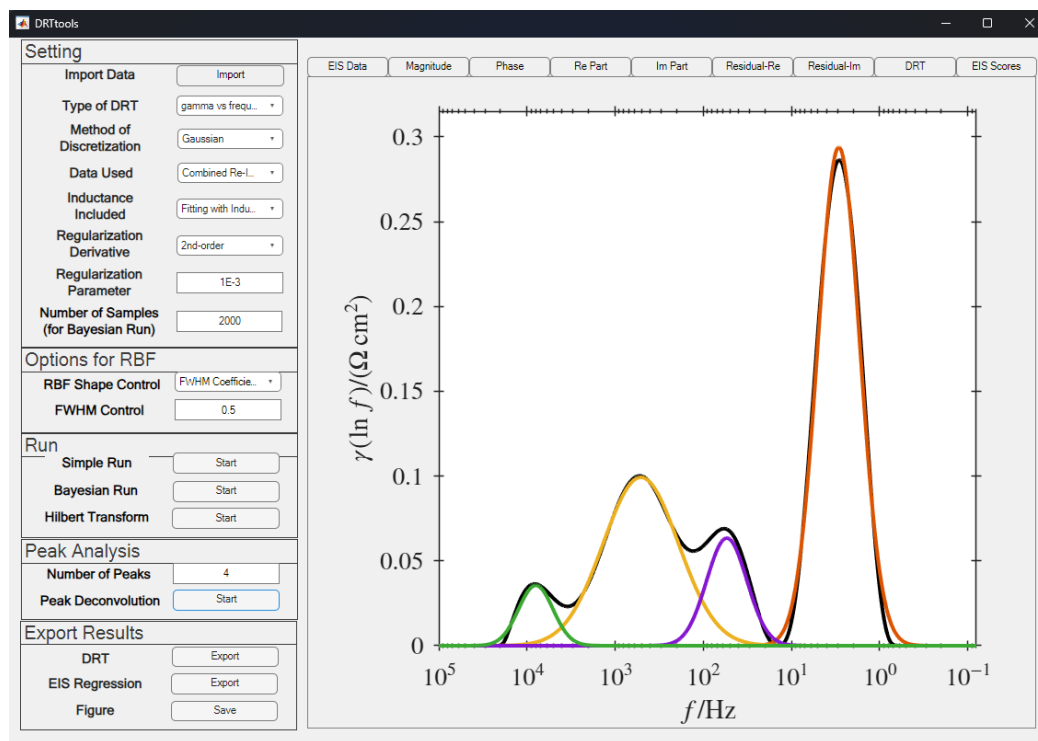
8. Select the number of peaks visible in the DRT plot.



9. Click “Start Peak Deconvolution” to perform peak deconvolution.

In some cases, you may need to run “Simple Run” again before starting the peak deconvolution.

10. Click “Save Figure” to save the resulting figure and a CSV file including polarization resistance (R), capacitance, FWHM (Full Width at Half Maximum), and peak frequency/relaxation time (τ) for each peak of the DRT curve.



Solartron figure.fig	2026-08-15 9:27 PM	MATLAB Figure	116 KB
Solartron parameters.csv	2026-08-15 9:27 PM	Microsoft Excel C...	1 KB

A	B	C	D	E	F	G	H
ID	Series_Name	Area_Ohm_cm2	R_Ohm_cm2	Tau_s	Capacitance_F_per_cm2	Frequency_Hz	FWHM_InTau
1	data5	0.040087503	0.039587731	2.00E-05	0.0005056	7951.560002	1.057878657
2	data3	0.244863488	0.241810764	0.000309687	0.0012807	513.9218011	2.317528829
3	data4	0.083636292	0.082593594	0.002929681	0.035471049	54.32499972	1.238706207
4	data2	0.402817714	0.397795767	0.054603347	0.137264776	2.914747037	1.287718878
5	Total Sum	0.771404997	0.761787855	NaN	NaN	NaN	NaN

Contact and Citation

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Masood Fakouri Hasanabadi

Email: fakourih@ualberta.ca

If you use this software in your research, please cite the GitHub repository using the “Cite this repository” option available on GitHub:

GitHub: <https://github.com/MFakouri/DRT-Peak-Analysis-.git>

Please cite the software in any publications or presentations where it is used.